

Dr. Jialei Shi

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RESEARCH BRIEF

I am currently a Research Associate in Medical Robotics at Hamlyn Centre for Robotic Surgery, Imperial College London. I specialise in **design, modelling and control of mechatronics and robotics systems**, in particular, **soft robotics for healthcare applications**. My work spans from **soft flexible medical robotics** to other **complex mechatronics systems**, e.g., haptics interface. My vision is to create and apply **innovative soft robotic systems** for addressing translational challenges.

HIGHLIGHTS AND ACHIEVEMENTS

- **Full Ph.D scholarship in Soft Robotics** at Univeristy College London (UCL, QS2026 9th). More than eight years of both **hardware and software experiences** in **design, modelling, control and validation of mechatronic and soft robotic systems**. I have solid **hands-on and mathematical skills of, CAD design, electronics, micro-controller programming design and fabrication of soft robots, system integration, analytical modelling and simulation, and control implementation**.
- Top and intelligent student with distinct grades, e.g., ranking **1st (1/110)** in the four-year undergraduate project. A **recipient of multiple competitive institutional, national and international awards and scholarships**, e.g., the UCL Dean's Prize, CSC-UCL Joint Research Scholarship, UK Turing Scheme.
- **Strong academic publication/presentation track record and writing abilities**. (Co)authored over 15 publications on peer-reviewed, high-impact journals and prestigious conferences, e.g., The International Journal of Robotics Research (**IJRR, Impact Factor: 7.5**), IEEE Transactions on Robotics (**TRO, Impact Factor: 9.4**), IEEE/ASME Transactions on Mechatronics (**T-Mech, Impact Factor: 6.1**), ICRA and IROS (Tier 1 robotics conferences), and RoboSoft (Tier 1 soft robotics conference).
- Recipient of **RSS Pioneers 2024 in the prestigious conference Robotics: Science and System (30 pioneers worldwide per year, acceptance rate < 15%)**. The first ever selected representative from the UCL.
- Excellent **communication, teamwork management, and collaboration abilities**, e.g., founded and led the first autonomous racing team with over 50 team members; **collaborated and worked with researchers/clinicians** from different **departments within UCL or other external institutes**, e.g., University of Cambridge (UK), King's College London (UK), Inria Centre (France), SUST (China).
- **Extensive teaching (awarded as Associate Fellow by Advanced HE) and supervision experiences**, e.g., acted as a postgraduate teaching assistant to design lecture materials, deliver online/on-site tutorials, workshops and lab demonstrations for undergraduates and masters.
- **Reviewer** for the IJRR, IEEE TRO, IEEE T-MECH, IEEE RA-L, ICRA, IROS, RoboSoft, Humanoids, awarded as **2025 IEEE Robotics and Automation Letters (RA-L) Outstanding Reviewer**; Co-organised ICRA/IROS Tutorial/Workshop.

EDUCATION

University College London (9th world rank QS 2025) <i>Mechanical Engineering (Full scholarship)</i>	Ph.D Student 09. 2019 – 02. 2024
Tsinghua University (Top 2 in China) <i>Automobile Engineering</i>	Visiting Student 12. 2017 – 06. 2019
Beijing Institute of Technology (Top 10 in China) <i>Aerospace Engineering (Distinct)</i>	Master 09. 2017 – 06. 2019
Harbin Institute of Technology (Top 10 in China) <i>Naval Architecture and Ocean Engineering (First-Class Honours, 1/110 students)</i>	Bachelor 09. 2013 – 06. 2017

RESEARCH EMPLOYMENT

Imperial College London <i>Hamlyn Centre for Robotic Surgery, Mechanical Engineering Department</i>	Research Associate 03. 2024 – present
Imperial College London <i>Hamlyn Centre for Robotic Surgery, Mechanical Engineering Department</i>	Research Assistant 12. 2023 – 02. 2024
University College London <i>Mechanical Engineering Department</i>	Research Assistant 12. 2022 – 11. 2023

RESEARCH PUBLICATIONS

1. Journal Publications:

1.1 Published:

- J1. **J. Shi**, K. Borvorntanajanya, K. Chen, E. Franco, F.B. Baena. Design, control and evaluation of a novel soft evertng robot for colonoscopy, *IEEE/ASME Transactions on Robotics*, Vol. 4(1), pp. 4843-4859, 2025. [\[video\]](#) (**IF=10.5**)
- J2. J. F. Ahmed, Borvorntanajanya, K. **J. Shi**, E. Franco, A. Darzi, F.B. Baena, N. Patel. Reducing applied force in colonoscopy using a novel soft robotic colonoscope: A head-to-head study, *Endoscopy International Open*, pp. 1-7, 2025. (**IF=2.6**)
- J3. K. Borvorntanajanya, **J. Shi**, J. F. Ahmed, E. Franco, F.B. Baena. A pneumatic force sensor for enhanced force feedback in robotic colonoscopy, *IEEE Transactions on Medical Robotics and Bionics*, pp.1-12, 2025. (**IF=3.8**)
- J4. **J. Shi**, S.-A. Abad, G. Shi, W. Gaozhang, J.S. Dai, H.A. Wurdemann. Model-based static compliance analysis and control for pneumatic-driven soft robots, *IEEE/ASME Transactions on Mechatronics*, pp. 1-12, 2025. [\[video\]](#) (**IF=7.3**)
- J5. G. Shi, **J. Shi**, A. Shariati, K. Motaghedolhagh, S. Homer-Vanniasinkam and H. A. Wurdemann, "Multi-cavity, fluidic haptic feedback system for mechano-tactile feedback," *IEEE Transactions on Haptics*, Vol. 18(1), pp. 6-19, 2025. (**IF=2.8**)
- J6. **J. Shi**, G. Shi, Y. Wu, H. A. Wurdemann. A multi-cavity touch interface for a flexible soft laparoscopy device: Design and evaluation, *IEEE Transactions on Medical Robotics and Bionics*, Vol. 6(4), pp.1309-1321, 2024. [\[video\]](#) (**IF=3.8**)

- J7. **J. Shi**, S.-A. Abad, J.S. Dai, H. A. Wurdemann. Position and orientation control for hyperelastic multi-segment continuum robots, *IEEE/ASME Transactions on Mechatronics*, Vol. 29(2), pp. 995-1006, 2024. [\[video\]](#) (**IF=5.0**)
- J8. **J. Shi**, J. Shi, A. Shariati, S.-A. Abad, Y. Liu, J.S. Dai, H. A. Wurdemann. Stiffness modelling and analysis of soft fluidic-driven robots using Lie theory, *The International Journal of Robotics Research*, vol. 43(3), pp. 354-384, 2024. [\[video\]](#) (**IF=5.0**)
- J9. Y. Gao, A. Redfearn, S. Dawes, **J. Shi**, H. A. Wurdemann, A. Taylor, S. Guldin, "A smart centrifuge for automated sample processing with liquid handling robots", pp. 1-14, *Journal of Open Hardware*, 2024. (**IF=2.2**)
- J10. W. Gaozhang, Y. Li, **J. Shi**, Y. Wang, A. Stilli and H. A Wurdemann, "A novel stiffness-controllable joint using antagonistic actuation principles," *Mechanisms and Machine Theory*, vol. 196, pp. 105614, 2024. [\[video\]](#) (**IF=5.3**)
- J11. R. M. Monteiro, **J. Shi**, H. A. Wurdemann, G. T. Thuruthel, Visuo-dynamic self-modelling of soft robotic systems, *Frontiers in Robotics and AI*, 2024. [\[video\]](#) (**IF=3.0**)
- J12. E. Almanzor, F. Ye, **J. Shi**, G. T. Thuruthel, H. Wurdemann, F. Iida. Static shape control of soft continuum robots using deep visual inverse kinematic models. *IEEE Transactions on Robotics*, vol. 39(4), pp. 2973-2988, 2023. [\[video\]](#) (**IF=10.3**)
- J13. L. Liang, X. Ping, **J. Shi**, X. Wang and X. Wu. Energy recovery strategy for regenerative braking system of intelligent four-wheel independent drive electric vehicles. *IET Intelligent Transport Systems*, vol. 15(1), pp. 119-131, 2021. (**IF=2.3**)
- J14. W. Wang, Y. Wu, **J. Shi**, L. Liang. Electronic hydraulic brake power system control strategy based on driver intention recognition, *Journal of Jilin University (Engineering and Technology Edition)*, 51(2), pp. 406-413, 2021.
- J15. X. Wang, X. Wu, S. Cheng, **J. Shi**, X. Ping and W. Yue. Design and experiment of control architecture and adaptive dual-loop controller for brake-by-wire system with an electric booster. *IEEE Transactions on Transportation Electrification*, vol. 6(3), pp. 1236-1252, 2020. (**IF=8.3**)
- J16. **J. Shi**, L. Li, X. Wang and C. Liu. Robust output feedback controller with high-gain observer for automatic clutch. *Mechanical Systems and Signal Processing*, vol. 132, pp. 806-822, 2019. (**IF=8.9**)
- J17. X. Wang, W. Wang, L. Li, **J. Shi** and B. Xie. Adaptive control of DC motor servo system with application to vehicle active steering. *IEEE/ASME Transactions on Mechatronics*, vol. 24(3), pp. 1054-1063, 2019. (**IF=7.3**)
- J18. Y. Sun, W. Wang, L. Li, **J. Shi** and Q. An. Modelling and control for economy-oriented car-following problem of hybrid electric vehicle. *IET Intelligent Transport Systems*, vol. 13(5), pp. 825-833, 2019. (**IF=2.3**)

1.2 In Review:

- J19. **J. Shi**, S.-A. Abad, J.S. Dai, H.A. Wurdemann. Compliance model-based contact force control for soft continuum robots, *Soft Robotics* (revised and resubmitted).
- J20. **J. Shi**, H. Jin, S.-A. Abad, W. Gaozhang, G. Shi, H.A. Wurdemann. A static modelling and evaluation framework for soft robots with reinforced chambers, *IEEE Transactions on Robotics* (revised and resubmitted).

2. International Conference Papers:

- C1. Y. Huang, A. Alkayas, **J. Shi**, F. Rendo, H. A. Wurdemann, T. Thuruthel. Predicting interaction shape of soft continuum robots using deep visual models, in *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, pp. 11381-11387, 2024.
- C2. **J. Shi**, A.-A. Abad, A. Menciassi, K. Althoefer, H. A. Wurdemann, Miniaturised soft manipulators with reinforced actuation chambers on the sub-centimetre scale, in *IEEE 7th International Conference on Soft Robotics (RoboSoft)*, pp. 157-164, 2024. [[video](#)]
- C3. X. Chen, **J. Shi**, H. A. Wurdemann, G. T. Thuruthel, Vision-based tip force estimation on a soft continuum robot, in *IEEE International Conference on Robotics and Automation (ICRA)*, pp. 7621-7627, 2024.
- C4. **J. Shi**, H. A. Wurdemann, Design and control of a tele-operated soft instrument in minimally invasive surgery, in *Hamlyn Symposium on Medical Robotics*, pp. 31-32, 2023.
- C5. W. Gaozhang, **J. Shi**, Y. Li, A. Stilli and H. A. Wurdemann, Characterisation of antagonistically actuated, stiffness-controllable joint-link units for cobots, in *IEEE International Conference on Robotics and Automation (ICRA)*, pp. 655-661, 2023. [[video](#)]
- C6. **J. Shi**, W. Gaozhang, H. Jin, G. Shi and H. A. Wurdemann, Characterisation and control platform for pneumatically driven soft robots: Design and applications, *IEEE 6th International Conference on Soft Robotics (RoboSoft)*, pp. 1-8, 2023. [[video](#)]
- C7. G. Shi, A. Shariati, **J. Shi**, N. Herzig, S.A. Abad, H. A. Wurdemann, Modelling the inflation of an elastic membrane with a load, *IEEE 6th International Conference on Soft Robotics (RoboSoft)*, pp. 1-8, 2023.
- C8. P. Chaillou*, **J. Shi***, A. Kruszewski, I. Fournier, H. A. Wurdemann, C. Duriez, Reduced finite element modelling and closed-loop control of pneumatic-driven soft continuum robots, *IEEE 6th International Conference on Soft Robotics (RoboSoft)*, pp. 1-8, 2023. (equal contribution *) [[video](#)]
- C9. **J. Shi**, W. Gaozhang and H. A. Wurdemann, Design and characterisation of cross-sectional geometries for soft robotic manipulators with fibre-reinforced chambers, *IEEE 5th International Conference on Soft Robotics (RoboSoft)*, pp. 125-131, 2022. [[video](#)]
- C10. **J. Shi**, J. C. Frantz, A. Shariati, A. Shiva, J. S. Dai, D. Martins and H. A. Wurdemann. Screw theory-based stiffness analysis for a fluidic-driven soft robotic manipulator. in *IEEE International Conference on Robotics and Automation (ICRA)*, pp. 11938-11944, 2021. [[video](#)]

- C11. A. Shariati, **J. Shi**, S. Spurgeon and H. A. Wurdemann. Dynamic modelling and visco-elastic parameter identification of a fibre-reinforced soft fluidic elastomer manipulator. in *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2021. [\[video\]](#)
- C12. **J. Shi**, C. Huang, X. Wang and L. Li. Adaptive dual-loop hydraulic pressure controller for electric booster brake system. in *IEEE 28th International Symposium on Industrial Electronics (ISIE)*, pp. 1481-1486, 2019.
- C13. C. Huang, L. Li, **J. Shi** and X. Wang. Decoupling Control of a SISO LTI Cascaded System with Inner Feedback Loops. in *IECON 2019 - 45th Annual Conference of the IEEE Industrial Electronics Society*, pp. 341-346, 2019.

3. Patent:

- P1. **J. Shi**, F. Rodriguez. A Self-propelled Robotic Endoscope, GB9015935, 2025. (filed)

HONOURS AND AWARDS

National and International Awards

IEEE Robotics and Automation Letters (RA-L) Outstanding Reviewer	05. 2025
<i>Awarded for the distinguished service and commitment for the IEEE RA-L, 10 recipients worldwide per year</i>	
RSS Pioneers	07. 2024
<i>RSS Pioneers is with the Robotics: Science and Systems (RSS) conference, 30 recipients worldwide per year</i>	
Turing Scheme Grant	04. 2023
<i>The UK's global programme to support students to study and work aboard, funded by the UK Government</i>	
Best MathWorks' Poster Prize	04. 2023
<i>The best poster prize awarded by the MathWorks at the IEEE 2023 RoboSoft conference</i>	
China Scholarship Council-UCL Joint Research Scholarship	06. 2019
<i>Totally awarding 25 Chinese PhD students per year across UCL</i>	
Provincial Excellent Graduates Award	06. 2017
<i>Top 2%, for outstanding undergraduate performance.</i>	
The Second Prize of Ministry of Industry and Information Technology of China	03. 2017
<i>Team member</i>	
The National Championship of 2016 Formula Student Electric China	11. 2016
<i>Team leader</i>	
The Best Electrical System Design Award of Student Formula Japan	09. 2016
<i>Team leader</i>	

Institutional Awards

UCL Highly Commended Thesis Award	06.2025
<i>Awarded by the UCL Mechanical Engineering Department</i>	
UCL Mechanical Engineering Conference Award	02. 2022 - 02. 2023
<i>Three times, awarded by the UCL Mechanical Engineering Department</i>	
The Runner-up Poster Prize	02. 2023
<i>Awarded by the UCL Robotics Institute</i>	

UCL Dean's Prize	01. 2019
<i>1 PhD student per year in the Mechanical Engineering Department, award by the UCL</i>	
The First Prize Scholarship of Postgraduate	10. 2017
<i>Top 5%, for outstanding academic performances</i>	
Candidate of Top10 Excellent Undergraduate Students	04. 2017
<i>Top 0.15%, for undergraduate performances, award 20 students per year across the whole university</i>	
The First Prize Scholarship of Undergraduate	2014-2015
<i>Top 5%, for outstanding academic performances</i>	

RESEARCH EXPERIENCES

Self-propelled Soft Robotic Endoscopes for Gastrointestinal Surgery 12. 2023 - present
Imperial College London, UK (Research Assistant/Associate)

- **Designed and manufactured an soft everting robot with an integrated flexible steering tip devices to advance the endoscopy technology.**

Static Modelling and Control for Pneumatic-driven Soft Continuum Robots 09. 2019 - 02. 2024
University College London, UK (Ph.D Student)

- **Designed, manufactured, and characterised soft manipulators with fully fibre-reinforced chambers on different scales for surgical applications.** For the first time, the **achieved diameter of these robots is less than 10 mm** [[video](#)].
- Developed and validated **mathematical models** to describe the **forward kinematics, tip force generation and configuration-dependent compliance** of soft robots.
- Developed and validated **model-based control algorithms** for soft robots, to achieve **inverse kinematics control, on-demand force control, and compliance control**.
- **Transferred knowledge of soft robot fabrication and characterisation**, resulting in collaborations with colleagues to develop **soft material-based variable stiffness joints and haptic feedback devices**.

Development of a Pneumatic-driven Soft Haptic Fingertip Device 11. 2022 - 11.2023
University College London, UK (Research Assistant)

- **Worked with clinical neurophysiologists** on the SoftSensation Project (UCL Therapeutic Acceleration Support Fund, PI: Dr. Sara Abad Guaman) to create **pneumatic-actuated, soft material-based haptic fingertip device** to characterise and restore the sense of touch of human's fingertip.
- Developed **software interface and micro-controller (Teensy 4.1) programming** for the haptic device, including a Python interface which communicates with programmed micro-controllers using C.

Design and Control of Soft Instrument for Cancer Imaging 06. 2023 - 09. 2023
Inria, University of Lille, France (Visiting Researcher)

- Supported by the **UK Turing Scheme**, aiming at **acquiescing 3D imaging scanning for cancer diagnosis**, a 11.5 mm soft instrument made of two robotic segments was successfully developed, which can seamlessly fit to commercial 12 mm trocar ports.
- Collaborated with researchers from DEFROST lab led by Prof. Christian Duriez, to develop **reduced finite element modelling and closed-loop control** of the soft instrument using SOFA [[video](#)].

Brake-by-wire System Design and Control for Electric Vehicles

01. 2018 - 06. 2019

Tsinghua University, China (M.S. Student)

- Designed an **electro-hydraulic brake-by-wire system of electric vehicles**, including **mechanical structure design and programming**. The system is capable of realising brake-assistant function and coordinate with traction motor to optimize energy recovery efficiency of vehicles.
- Developed a **robust and adaptive back-stepping controller**, which can control clutch position of Hybrid Electric Vehicles (HEVs) with high accuracy when un-modelled uncertainties present, validated by **numerical simulations using Matlab/Simulink and experiments**.

NIO Autonomous R&D Department

12. 2016 - 04. 2017

NIO Co.,Ltd, China (Intern)

- Designed a **multi-sensor fusion platform for testing algorithms of Advanced Driver Assistance Systems (ADAS) on road conditions**.
- Studied and optimised **lane change algorithms** of vehicles using polynomial and verified the proposed algorithm by **model-in-loop (MIL)**.

Formula Student Electric China (FSEC) & Student Formula Japan (SFJ)

12. 2014 - 09. 2017

Harbin Institute of Technology, China (Bachelor Student)

- Devoted over three years in the ([HIT Racing Team](#)). and participated **4 times Formula Student Electric China, i.e., FSEC** (Shanghai, China) and **3 times Student Formula Japan, i.e., SFJ** (Shizuoka, Japan). Focused on the **power-train and electronics system design** of electric racing cars.
- As a subsystem leader in 2015, team won overall the second place in FSEC. As a **team leader** (with over 60 students) and driver in 2016, **led team to win totally 12 awards with more than 400,000 Chinese Yuan, and the national championship in 2016 FSEC**. In addition, led team to win **Best Electrical System Design Awards in 2016 SFJ**.
- **Founded the first autonomous racing team in Harbin Institute of Technology**, with more than 30 students. Led the team to win **overall the second place in Formula Student China 2017**.

TALKS AND PRESENTATIONS

- **Invited talk:** Integrated framework and benchmarking platform for characterisation, control, and modelling of soft robots, *IEEE RoboSoft Workshop* on benchmarking in soft robots, April, 2025.
- **Oral presentation:** Understanding compliance properties of soft continuum robots: from analytical model to model-based control, *Robotics: Science and Systems (RSS) Pioneers*, June, 2024.
- **Invited talk:** Characterisation and control platform for pneumatically driven soft robots, *IEEE ICRA tutorial* on towards an accessible soft robotics toolbox and validation test rig, June, 2023.
- **Oral presentation:** Characterisation of miniaturised soft continuum robots with reinforced chambers, *The Third International Nonlinear Dynamics Conference*, June, 2023.
- **Oral presentation:** Characterisation and control platform for pneumatically driven soft robots: Design and applications. *IEEE RoboSoft*, April, 2023.
- **Oral presentation:** Design, modelling and control for soft continuum robots with full chamber reinforcement. *UCL Robotics*, February, 2023. ([Recording](#)).
- **Invited talk:** Design and modelling for chamber-Reinforced soft robots with dimension scalability, *Exeter International Symposium on Small-Scale Robots*, June, 2022. ([Recording](#)).
- **Oral presentation:** Design and characterisation of cross-sectional geometries for soft robotic manipulators with fibre-reinforced chambers. *IEEE RoboSoft*, April, 2022. ([Recording](#)).

PROFESSIONAL HISTORY

POSTGRADUATE TEACHING ASSISTANT (PGTA)

11. 2019 - 12. 2023

University College London, UK

- Supported **teaching activities for undergraduates and masters** across several aspects, including lab supervision, assessment marking, and tutorial delivery, e.g., for Mathematical Modelling and Analysis, Control and Instrumentation, Robotics and Control Modules.
- **Supervised one undergraduate student** (admitted by Carnegie Mellon University, US) to develop soft robotics toolkits (see publication C4) and **Co-supervised two Master students to achieve learning-based force estimation of soft robots**, resulting in an IROS (C1) and an ICRA paper (C3).
- **Awarded as Associate Fellow under the Professional Standards Framework by [Advanced HE](#)** (Certificate no. PR263476).

Workshop Organisation and Reviewer Roles

- **Tutorial Co-organiser at IEEE ICRA 2023:** Towards an accessible soft robotics toolbox and validation test rig. [link](#)
- **Conference Paper Reviewer:** IEEE ICRA, IROS, RoboSoft, Humanoids - since 2019.
- **Journal Paper Reviewer:** International Journal of Robotics Research, IEEE Transactions on Robotics, IEEE Robotics and Automation Letters, Mechanism and Machine Theory - since 2018.

Competition Events

- **Event Coordinator of Formula Student China:** Volunteered to coordinate the technical scrutineering for more than 20 formula student teams, October, 2018.
- **Design Judge of Formula Student Switzerland:**Volunteered to be the design judge to evaluate the engineering design of electric racing cars, July, 2023.

SKILLS SUMMARY

Hardware Design and Development

- Proficient in **CAD design, structure analysis, machining and 3D printing.**
- Solid **hand-on skills for building, assembly, integration and testing of mechatronics systems.**
- Proficient in design, fabrication, test and characterisation of **silicone-made soft robots.**
- Knowledgeable of **electronics system development, test and debugging.**

Software Skills

- Experienced in using **C/C++, Matlab/Simulink, Robot Operating System (ROS), Python,** to achieve control implementation, graphical user interface development and micro-controller programming.
- **Deep understanding of control system;** experienced in **deploying and simulating various control algorithms,** from PID to adaptive control.
- Proficient in **solving mathematical or optimisation problems..**